

Signal Analysis & Telemetry Extraction Report

Target Recording: TEST_SAT_QPSK_DEGRADED.iq | Format: IQ | Sample Rate: 500.0 kSps

SIGNAL HEALTH
38/100
STATUS: CRITICAL

ANALYST PRIORITY
HIGH
Critical parameter
anomal...

CONFIDENCE
90.6%
DSP DATA QUALITY

ANOMALIES
3
DETECTED ISSUES

1. Explainable Analysis & Executive Summary

Automated analysis completed for 'TEST_SAT_QPSK_DEGRADED.iq'.

The recording contains 1 detected active segment(s) across a total duration of 0.120s. Spectral analysis identifies a center frequency at 88.490 kHz with an occupied bandwidth of 463.379 kHz and an SNR of 23.7 dB (Modulation profile: Complex Telemetry / Multi-carrier).

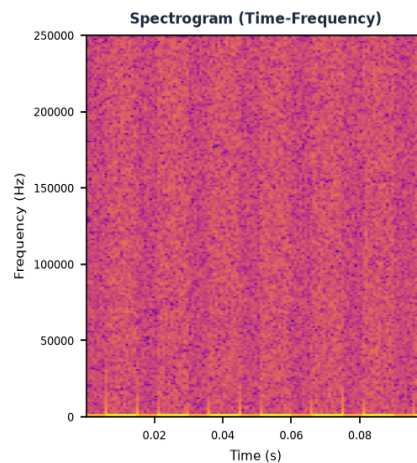
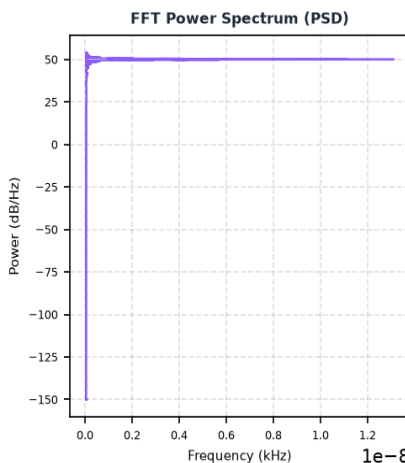
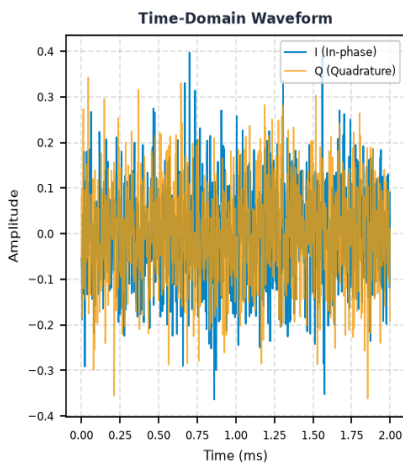
Benchmarked against historical baseline 'SAT-QPSK-REF-BASELINE'. Statistical drift analysis indicates baseline center frequency was 80037.0 Hz (observed: 88489.9 Hz) and nominal SNR was 48.2 dB.

Detected 3 anomaly condition(s):

- Center Frequency (CRITICAL): Center frequency drifted by +10.6% (8453.0 Hz) from baseline mean (5.3 σ deviation). Possible oscillator frequency drift or Doppler shift.
- Signal-to-Noise Ratio (SNR) (CRITICAL): SNR degraded by 24.5 dB below historical baseline. Indicates elevated channel noise floor, attenuation, or RF interference.
- Occupied Bandwidth (HIGH): Occupied bandwidth expanded/compressed by +1305.9% from nominal baseline. Indicates symbol rate change, non-linear distortion, or spectral regrowth.

Overall Signal Health Score is 38.0/100 (CRITICAL), assigning this file to HIGH PRIORITY for engineer triage.

2. Time-Domain, Spectral & Time-Frequency Visualization



3. Extracted Signal Parameters

MEASURED PARAMETER	VALUE	DERIVED / ESTIMATED PARAMETER	VALUE
Peak Frequency	103.516 kHz	Occupied Bandwidth (99% OBW)	463.379 kHz
Spectral Centroid	88.490 kHz	-3 dB Bandwidth	13.184 kHz

Signal Power (dBFS)	-6.84 dB	-10 dB Bandwidth	28.809 kHz
Est. Noise Floor	-73.1 dB	Signal Stability Index	0.745 / 1.0
Measured SNR	23.67 dB	Crest Factor / PAPR	2.199 (6.84 dB)
Peak Amplitude	1.0 V	Spectral Flatness (Entropy)	0.2197
RMS Amplitude	0.4547 V	Modulation Classifier Hint	Complex Telemetry / Multi-carrier (75.0%)

4. Signal Fingerprint & 'What Changed?' Analysis (Fingerprint ID: FP-BDCCB2035645D769)

PARAMETER	BASELINE MEAN	CURRENT VALUE	DELTA (%)	DIR	STATUS
Center Frequency	80036.97 Hz	88489.95 Hz	+10.6%	↑	Anomalous High
Bandwidth (99% OBW)	32958.98 Hz	463378.91 Hz	+1305.9%	↑	Anomalous High
Signal-to-Noise Ratio (SNR)	48.22 dB	23.67 dB	-24.5%	↓	Anomalous Low
Signal Power	-2.27 dBFS	-6.84 dBFS	-4.6%	→	Nominal
Signal Stability	0.913 index	0.745 index	-18.4%	↓	Anomalous Low
RMS Amplitude	0.77 V	0.455 V	-40.9%	↓	Anomalous Low

5. Anomaly Investigation Log (3 events)

SEVERITY	AFFECTED PARAMETER	OBSERVED VALUE	ENGINEERING REASON
CRITICAL	Center Frequency	88489.9 Hz	Center frequency drifted by +10.6% (8453.0 Hz) from baseline mean (5.3σ deviation). Possible oscillator frequency drift or Doppler shift.
CRITICAL	Signal-to-Noise Ratio (SNR)	23.7 dB	SNR degraded by 24.5 dB below historical baseline. Indicates elevated channel noise floor, attenuation, or RF interference.
HIGH	Occupied Bandwidth	463378.9 Hz	Occupied bandwidth expanded/compressed by +1305.9% from nominal baseline. Indicates symbol rate change, non-linear distortion, or spectral regrowth.